

CSCE 658 RANDOMIZED ALGORITHMS · MODULE 02 · LECTURE NOTES

Concentration

The whole course in one module.

MODULE OUTCOMES

- Apply Markov's inequality and know its weakness.
- Apply Chebyshev and explain what the variance buys.
- Apply Chernoff bounds and state the independence requirement.
- Apply the union bound and recognise when it is wasteful.
- Choose the right inequality for a given situation.

1 The ladder

Markov: $P(X \geq a) \leq E[X] / a \quad (X \geq 0)$

Chebyshev: $P(|X - \mu| \geq k\sigma) \leq 1/k^2$

Chernoff: $P(X \geq (1+\delta)\mu) \leq \exp(-\delta^2\mu/3)$

Union bound: $P(A_1 \cup \dots \cup A_n) \leq \sum P(A_i)$

Each step up the ladder assumes more and gives more. Markov needs only non-negativity and the mean; Chebyshev needs the variance; Chernoff needs independence. **And the jump from polynomial decay (Chebyshev's $1/k^2$) to exponential decay (Chernoff's $e^{-\delta^2\mu/3}$) is where randomised algorithms become practical** — a polynomial tail bound rarely gives a usable failure probability and an exponential one almost always does. **Presenting these as a ladder rather than as four facts is the right framing**, because the question in practice is always 'how much can I assume, and therefore which rung am I on'.

MARKOV IS WEAK, ALWAYS APPLIES, AND IS THE SOURCE OF THE OTHERS

$P(X \geq a) \leq E[X]/a$, for any non-negative X . That is all it needs — no variance, no independence, no distributional assumption of any kind.

And it is correspondingly weak: it gives only $1/k$ decay, so the probability of exceeding ten times the mean is bounded by $1/10$ — **which is usually a vast overestimate**, and is frequently a useless bound in practice.

But Chebyshev is Markov applied to the non-negative variable $(X - \mu)^2$, whose expectation is the variance; **and Chernoff is Markov applied to e^{tX}** , optimised over t . **So the whole ladder is Markov with a cleverly chosen function.**

Which is worth knowing because it tells you how to derive a bound you have not memorised: find a non-negative function of your variable whose expectation you can compute, apply Markov to it, and translate back. That procedure is the actual technique, and the three named inequalities are three instances of it.

2 Chernoff

FOR $X =$ a sum of n INDEPENDENT indicator variables, with mean μ :

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upper tail:  $P(X \geq (1+d)\mu) \leq \exp(-d^2 \mu / 3)$ 
              for  $0 < d \leq 1$ 
lower tail:  $P(X \leq (1-d)\mu) \leq \exp(-d^2 \mu / 2)$ 
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THE EXPONENT CONTAINS μ , which is the whole point: the MORE trials, the TIGHTER the concentration. A sum of a million fair coin flips lies within 0.5% of its mean with probability about $1 - 2e^{-8}$.

WHAT IT REQUIRES: INDEPENDENCE. State it every time. The bound is simply FALSE for dependent variables, and applying it to variables that are not independent is the commonest error in this subject.

WHAT TO DO WHEN THEY ARE NOT INDEPENDENT

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NEGATIVE ASSOCIATION -- Chernoff still holds. Balls
    thrown into bins has this property (Module 03).
MARTINGALES / Azuma -- bounded differences along a
    process; covers many dependent cases naturally.
McDIARMID -- a function of independent variables that
    changes by at most  $c_i$  when input  $i$  changes.
OR Chebyshev, which needs only PAIRWISE independence
    and is much weaker (and is exactly what Module 12's
    derandomisation exploits).
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Independence is the assumption to state out loud, every time — and the three alternatives above cover most of the cases where it fails. McDiarmid in particular is underused: a great many quantities

are functions of independent inputs that change only a little when one input changes, and that is enough for an exponential bound without the quantity being a sum at all.

3 The union bound

THE UNION BOUND TURNS ONE BOUND INTO MANY, AT A LINEAR COST

$P(\text{some bad event happens}) \leq \text{the sum of the individual probabilities}$. No independence is needed, which is why it is used constantly and why it is the glue in almost every high-probability argument.

The standard pattern: bound one bin's overflow probability by $1/n^2$, union over the n bins, and conclude that no bin overflows with probability at least $1 - 1/n$ — which is how every 'with high probability, all of them are fine' result in this course is obtained (Module 03 §1 is the first instance).

So design the per-event bound to be $1/n^c$ for $c > 1$, because the union over n events then still leaves you with $1/n^{c-1}$ — and that is why 'with high probability' conventionally means $1 - 1/n^c$ in this literature, rather than being an arbitrary convention.

And it is wasteful when the events are strongly correlated — if ten thousand nearly identical events each have probability $1/1000$, the union bound gives 10, which is no bound at all. **The Lovász local lemma** (Module 07 §3) is the tool for exactly that situation, when the events are only *locally* dependent.

Situation	Use	Why
Only the mean is known	Markov	It is all you can do, and it always works.
Variance known, dependence unclear	Chebyshev	It needs only pairwise independence, which is far easier to establish — and is what makes Module 12's limited-independence derandomisation possible.
A sum of independent indicators	Chernoff	Exponential decay. Almost always this one, and see the note.
'All n of them are fine'	Union bound	Applied on top of whichever per-event bound you used.
A function of independent inputs with bounded differences	McDiarmid	The dependence is handled by the function's structure rather than needing to be removed.
Many locally dependent bad events	Lovász local lemma	When the union bound exceeds 1 and the dependencies are local (Module 07 §3).

The third row covers most of what you will actually do — and recognising a quantity as a sum of independent indicator variables is the skill, considerably more than knowing the inequality's statement. Most of Modules 03 through 05 consists of finding that framing for a problem that did not obviously have it.

4 Amplification

REPETITION CONVERTS A WEAK GUARANTEE INTO ANY GUARANTEE YOU WANT

One-sided error: repeat k times and accept only if all k runs accept. The failure probability becomes p^k — exponential in k , for linear cost, and no concentration bound is even needed.

Two-sided error: repeat and take the majority answer. A Chernoff bound on the number of correct runs gives exponential decay — which is CSCE 637 Module 08 §1's error amplification, derived here rather than asserted.

So a 1/3-error algorithm run a hundred times has failure probability below 2^{-30} , and the exponent is yours to choose by choosing k .

Which is the single most useful fact in this module: you do not have to accept the algorithm's natural error rate, and the cost of improving it is linear while the improvement is exponential. That trade is what makes Module 01 §3's hardware-reliability argument available, and it is the reason the subject's guarantees are engineering parameters rather than facts to live with.

- **Applying Chernoff without independence.** The commonest error by a wide margin, and the bound is simply false rather than merely loose — so check the assumption and name a fallback if it fails (§2's list).
- **Confusing the expectation with the tail.** An expected-time bound is not a latency guarantee (Module 01 §4, CSCE 678 Module 06) — and this is the error most likely to reach production.
- **Union-bounding over too many events.** If the per-event bound is $1/n$ and there are n events, you have proved nothing at all — the bound is 1, which is true of every probability.
- **Using Chebyshev where Chernoff applied.** This gives a polynomial bound where an exponential one was available, which makes a perfectly good algorithm look weak — and it happens because the independence was available and unnoticed.
- **And reporting the bound rather than measuring the actual rate.** Measure it, and compare against the bound (Module 13 §1) — the bound is usually loose, and reporting the measured rate alongside it is both more honest and more impressive. **The independence check is the one to make habitual:** write the assumption down before applying the bound, every time.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Mitzenmacher & Upfal — chapters 3–5	The ladder, derived — including the Markov-to-Chernoff derivation of §1's callout.

Resource	What it covers for this module
Motwani & Raghavan — chapters 3–4	The same material with more applications , and the martingale treatment of §2's fallbacks.
Boucheron, Lugosi & Massart — Concentration Inequalities	The comprehensive reference — McDiarmid, Azuma, and everything beyond. Library copy; the authors' survey papers are free.
Dubhashi & Panconesi — Concentration of Measure for the Analysis of Randomised Algorithms	§2's negative association and martingale material , aimed exactly at this course's needs. Library copy.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Derive Chebyshev from Markov** and Chernoff from Markov, showing the chosen function in each case.
- 2 **Compare all three bounds numerically** for a sum of 1000 fair coin flips at several deviations.
- 3 **Plot the three bounds and the true probability** on a log scale and report the gaps.
- 4 **Apply Chernoff to dependent variables deliberately** and construct a case where it fails.
- 5 **Find a negatively associated example** and confirm Chernoff still holds empirically.
- 6 **Apply McDiarmid to a function that is not a sum**, and verify the bound.
- 7 **Construct a union-bound argument** giving a $1 - 1/n$ guarantee from a per-event $1/n^2$ bound.
- 8 **Construct one where the union bound exceeds 1** and say what you would need instead.
- 9 **Amplify a 1/3-error algorithm** to below 10^{-15} and measure the empirical rate.
- 10 **Measure the empirical failure rate of one algorithm** and compare against your derived bound.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 State all four inequalities and what each assumes.
- 2 Derive Chebyshev and Chernoff from Markov.
- 3 Why is the jump to exponential decay the important one?
- 4 What does Chernoff require, and name four fallbacks.
- 5 Give the standard union-bound pattern and why $1/n^c$ is the target.
- 6 When is the union bound wasteful, and what replaces it?
- 7 Give the six-row lookup table.

- 8 Explain amplification for one-sided and two-sided error.
- 9 Name the four common errors and the habit that prevents the first.